KAIZEN MATHS

Lesson starter

Midpoint and Length of Line Segments

Missing Values

Name _____

S1

M (4, -2) is the midpoint of AB. A is (3, -4) . Find B.

S2

A is (0, 4) and B is (0, y) . AB = 5 units and B is below A. Find y.

S3

M (0, 2) is the midpoint of AB. A is (3, 4) . Find B.

S4

A is (1, -4) and B is (x, -4) . AB = 3 units and B is left of A. Find x.

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GEO-3384-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Missing Values

Name _____

E1

M (2, 1) is the midpoint of AB. A is (5, 5) . Find B.

E2

A is (1, -3) and B is (1, y) . AB = 3 units and B is above A. Find y.

E3

M (-2, 1) is the midpoint of AB. A is (-4, 0) . Find B.

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Missing Values

Name _____

E1

M (2, 1) is the midpoint of AB. A is (5, 5) . Find B.

E2

A is (1, -3) and B is (1, y) . AB = 3 units and B is above A. Find y.

E3

M (-2, 1) is the midpoint of AB. A is (-4, 0) . Find B.

GEO-3384-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Missing Values

Name _____

E1

M (2, 1) is the midpoint of AB. A is (5, 5) . Find B.

E2

A is (1, -3) and B is (1, y) . AB = 3 units and B is above A. Find y.

E3

M (-2, 1) is the midpoint of AB. A is (-4, 0) . Find B.

GEO-3384-L3 · E

Show your working

Teacher answers & working

Midpoint and Length of Line Segments · Missing Values

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S1

M (4, -2) is the midpoint of AB. A is (3, -4) . Find B.

Answer:

(5, 0)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(4) - (3) = 5$

$y_B = 2y_M - y_A = 2(-2) - (-4) = 0$

Therefore B = (5, 0).

S2

A is (0, 4) and B is (0, y) . AB = 5 units and B is below A. Find y.

Answer:

$y = -1$

Working:

The segment is vertical because the x-coordinate is unchanged.

$y = 4 - 5$

$y = -1$

Therefore the missing coordinate is y = -1.

S3

M (0, 2) is the midpoint of AB. A is (3, 4) . Find B.

Answer:

(-3, 0)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(0) - (3) = -3$

$y_B = 2y_M - y_A = 2(2) - (4) = 0$

Therefore B = (-3, 0).

Teacher answers & working

Midpoint and Length of Line Segments · Missing Values

GEO-3384-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

A is (1, -4) and B is (x, -4) . AB = 3 units and B is left of A. Find x.

Answer:

$x = -2$

Working:

The segment is horizontal because the y-coordinate is unchanged.

$x = 1 - 3$

$x = -2$

Therefore the missing coordinate is x = -2.

E1

M (2, 1) is the midpoint of AB. A is (5, 5) . Find B.

Answer:

(-1, -3)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(2) - (5) = -1$

$y_B = 2y_M - y_A = 2(1) - (5) = -3$

Therefore B = (-1, -3).

E2

A is (1, -3) and B is (1, y) . AB = 3 units and B is above A. Find y.

Answer:

$y = 0$

Working:

The segment is vertical because the x-coordinate is unchanged.

$y = -3 + 3$

$y = 0$

Therefore the missing coordinate is y = 0.

Teacher answers & working

Midpoint and Length of Line Segments · Missing Values

GEO-3384-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

M (-2, 1) is the midpoint of AB. A is (-4, 0) . Find B.

Answer:

(0, 2)

Working:

If M is the midpoint, each coordinate of M is halfway between A and B.

$x_B = 2x_M - x_A = 2(-2) - (-4) = 0$

$y_B = 2y_M - y_A = 2(1) - (0) = 2$

Therefore B = (0, 2).